

5. Understanding of Philosophy (Elective, 3 Credits)

A. General Information

Course Name	Understanding of Philosophy	Code	HA2209	Lecture-Activity -Credits (HW)	3-0-3(1)
Prerequisite Course	None				
Related Course	All				

B. Course Description

his course is intended to give students a general perspective on the philosophical, social and historical issues related to the science. It will introduce the main historical stream of scientific thinking from the ancient natural philosophy to the modern quantum revolution. Topics include the scientific method, cosmology, composition of matter, evolution, energy, cognitive science, artificial intelligence, information revolution, biotech revolution, the posthuman, and quantum revolution as well as the encounter of the East and the West in the field of science.

C. Course Objectives

- 1) To develop an understanding of the nature of scientific thinking and modern science
- 2) To understand the purpose and methodology of science, the structure and roles of scientific theories, and scientific explanations
- 3) To acquire perspectives and concepts suitable for critically analysing how science developed and how scientific knowledge is constructed, evaluated, and at times, radically altered
- 4) To understand how scientific concepts, laws, theories, and methods have historically been formed, changed, and evolved
- 5) To develop the ability to communicate clearly and concisely, using the written and spoken word, and to construct persuasive and adequately supported arguments

D. Texts

- 1) Adam Hart-Davis et al., eds. *Science: The Definitive Visual Guide*, Dorling Kindersley, 2009.* [HD]
- 2) BBC Documentary, *The Story of Science: Power, Proof and Passion*, 2011.* (6 parts) [S]
- 3) BBC Documentary, *Horizon: What Is One Degree?*, 2011.
- 4) Alex Rosenberg (2011). *Philosophy of Science: A Contemporary Introduction*,

3rd ed. Routledge.

- 5) Richard DeWitt (2010). *Worldviews: An Introduction to the History and Philosophy of Science*, 2nd ed., Wiley-Blackwell.
- 6) John Gribbin (2010). *Science: A History*, Penguin Books.
- 7) J.E. McClellan & H. Dorn (2006). *Science and Technology in World History: An Introduction*, 2nd ed. JHU Press.

E. Evaluation

Students are evaluated based on the Course Participation (50%) and the mid-term and final exams (20% each) as well as the attendance (10%). Course Participation is defined as a combination of in-class discussion and any form of contribution to productive class. Just talking is not contributing. Bringing in new ideas and information, listening to the opinion of others and arguing them with thoughtful insights of your own are what we seek.

<Grading Standards for Written Work>

- 1) Impressive, imaginative, exceeding my expectations, well-written, makes good use of assigned materials.
- 2) Good effort with good result. Answers all parts of the question.
- 3) Perfunctory; missed the point, did something well but not the assignment, or good but sloppy.
- 4) Bare minimum of effort. Consider yourself warned that work must improve.
- 5) Final grade will be +/- letter grades based on your performance across the assignments.

F. Course Policies

- 1) Students are expected to actively participate in class discussions, activities, and assignments.
- 2) Late assignments will receive a lowered grade. If you cannot turn in assignments on the due date, please let me know in advance.
- 3) You are asked to maintain open minds to the different opinions, and you are encouraged to place positive value on those differences. Points of view may be actively argued, but please respect others.
- 4) Make-up exams are possible only with my prior consent, and only in the most extreme circumstances.
- 5) Cheating and plagiarism will not be tolerated. If someone is caught cheating or plagiarizing, he or she will receive a grade of 0 for the test or assignment.